

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – EXPERIMENTS

|                  |                                                                                                          |               |                                   |
|------------------|----------------------------------------------------------------------------------------------------------|---------------|-----------------------------------|
| Course Code:     | CSA0612                                                                                                  | Course Title: | Design and Analysis of Algorithms |
| CO Assessed:     | CO3: Articulation<br>(BL4, BL5)                                                                          | Assessment:   | Assessment Tool 1 – Experiments   |
|                  |                                                                                                          | Total Marks:  | 100                               |
| CO3 Statement:   | Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication |               |                                   |
| Student Name:    | Shaik Azeem                                                                                              | Reg. No.:     | 192472307                         |
| Section / Batch: | CSE-AI                                                                                                   | Date:         |                                   |

### Instructions to Students

- Answer the following question. Total Marks: 100.
- Write the algorithm and execute the code for the given experiment.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

| Question Type | Focus Area                                                                                                                                                                                                                         | Questions |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Experiments   | Analyze how Quick Sort can be applied for efficient ranking. Identify issues such as pivot selection and data distribution. Apply divide and conquer techniques. Analyze performance and propose optimizations with justification. | Q1        |

### SECTION A – Experiments

#### Code of Conduct

I Shaik Azeem(192472307)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Shaik Azeem

### RUBRICS FOR CALCULATION

| Experiments |          |           |           |           |           |
|-------------|----------|-----------|-----------|-----------|-----------|
| Criteria    | Weightag | Level 4 – | Level 3 – | Level 2 – | Level 1 – |

|                                  | <b>e</b> | <b>Exemplary</b>                                                                                             | <b>Proficient</b>                                                       | <b>Developing</b>                                            | <b>Beginning</b>                                              |
|----------------------------------|----------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| Understand<br>ing of<br>Concepts | 25%      | Demonstrates<br>thorough<br>understanding and<br>effectively integrates<br>multiple<br>concepts/technologies | Good<br>understanding<br>with proper<br>integration of<br>most concepts | Basic<br>understanding;<br>limited or partial<br>integration | Poor<br>understanding<br>; unable to<br>integrate<br>concepts |
| Experi<br>mental<br>Design       | 30%      | Well-planned,<br>systematic<br>implementation with<br>optimal solution                                       | Correct<br>implementatio<br>n with minor<br>issues                      | Basic<br>implementation<br>with errors or<br>inefficiencies  | Incorrect or<br>incomplete<br>implementatio<br>n              |
| Use of<br>Tools                  | 20%      | Effective and<br>advanced use of<br>tools/technologies<br>with proper<br>configuration                       | Appropriate<br>use of tools<br>with correct<br>execution                | Limited or<br>inconsistent use<br>of tools                   | Incorrect or<br>minimal use<br>of tools                       |
| Analysis<br>of<br>Results        | 15%      | Insightful analysis<br>with accurate<br>interpretation and<br>validation of results                          | Good analysis<br>with<br>reasonable<br>interpretation                   | Basic analysis<br>with limited<br>insights                   | Poor or<br>incorrect<br>analysis of<br>results                |
| Documen<br>tation                | 10%      | Clear, well-structured<br>documentation with<br>diagrams, code, and<br>results                               | Organized<br>documentation<br>with necessary<br>details                 | Basic<br>documentation<br>with missing<br>details            | Poor or<br>incomplete<br>documentatio<br>n                    |

### Marks Evaluation:

| <b>Criteria</b>           | <b>Wt.</b> | <b>Level 4 –<br/>Exemplary</b> | <b>Level 3 –<br/>Proficient</b> | <b>Level 2 –<br/>Developing</b> | <b>Level 1 –<br/>Beginning</b> |
|---------------------------|------------|--------------------------------|---------------------------------|---------------------------------|--------------------------------|
| Understanding of Concepts | 25%        |                                |                                 |                                 |                                |
| Experimental Design       | 30%        |                                |                                 |                                 |                                |
| Use of Tools              | 20%        |                                |                                 |                                 |                                |
| Analysis of Results       | 15%        |                                |                                 |                                 |                                |
| Documentation             | 10%        |                                |                                 |                                 |                                |
| <b>Total Marks (100):</b> |            |                                |                                 |                                 |                                |

| <b>Faculty In-charge</b> | <b>Course Coordinator</b> | <b>Head of Department</b> |
|--------------------------|---------------------------|---------------------------|
| Name:                    | Name:                     | Name:                     |
| Signature: _____         | Signature: _____          | Signature: _____          |
| Date: _____              | Date: _____               | Date: _____               |

Q1

**Online Advertisement Ranking (Quick Sort)** An advertisement platform ranks ads based on relevance and bid value100  
marks

Analyze how Quick Sort can be applied for efficient ranking. Identify issues such as pivot selection and data distribution. Apply divide and conquer techniques. Analyze performance and propose optimizations with justification.

**Your Code**

```
1 ads = [  
2     ("Ad A",130),  
3     ("Ad B",150),  
4     ("Ad C",125),  
5     ("Ad D",150),  
6     ("Ad E",90),  
7     ("Ad F",145)  
8 ]  
9  
10 def quick_sort(arr):  
11     if len(arr)<=1:  
12         return arr  
13     pivot= arr[len(arr)//2][1]  
14  
15     left= [ad for ad in arr if ad[1]>pivot]  
16     middle= [ad for ad in arr if ad[1]==pivot]  
17     right= [ad for ad in arr if ad[1]<pivot]  
18  
19     return quick_sort(left) + middle + quick_sort(right)  
20 ranked_ads =quick_sort(ads)  
21  
22 print("Advertisement Ranking: ")  
23 for rank,(name, score) in enumerate(ranked_ads,1):  
24     print(f"{rank}. {name} - Score: {score}")
```

**Input**

stdin input

**Output**

```
Advertisement Ranking:  
1. Ad B - Score: 150  
2. Ad D - Score: 150  
3. Ad F - Score: 145  
4. Ad A - Score: 130  
5. Ad C - Score: 125
```

## Objective:

The objective is to rank online advertisements based on their relevance and bid value. Advertisements with higher scores should appear at the top so that the most valuable and relevant ads are displayed first.

## Problem Description:

An online advertising platform may receive a large number of advertisements. Each advertisement is assigned a ranking score, which can depend on factors such as:

- Bid amount
- Relevance to the user
- Advertisement quality
- User engagement

The advertisements need to be arranged in descending order of score.

## Why Quick Sort?

Quick Sort is suitable for advertisement ranking because it uses the divide-and-conquer technique.

The main idea is:

1. Select a pivot advertisement.
2. Divide the remaining advertisements into groups based on the pivot score.
3. Recursively sort the divided groups.
4. Combine the groups to obtain the final ranking.

For the given example, the advertisements have scores such as:

Advertisements Score

| Advertisement | Score |
|---------------|-------|
| Ad A          | 130   |
| Ad B          | 150   |
| Ad C          | 125   |
| Ad D          | 150   |
| Ad E          | 90    |
| Ad F          | 145   |

After sorting, the ranking is:

1. Ad B – 150
2. Ad D – 150
3. Ad F – 145
4. Ad A – 130
5. Ad C – 125
6. Ad E – 90

## Divide and Conquer Approach:

Quick Sort divides the advertisements into three groups relative to a selected pivot:

- Higher score than pivot
- Equal score to pivot
- Lower score than pivot

Each group is then processed recursively until the advertisements are completely ordered.

## **Pivot Selection:**

Pivot selection has an important effect on Quick Sort's performance.

- Good pivot: Divides the data into reasonably balanced groups.
- Poor pivot: Creates highly unbalanced groups.
- A middle element, random element, or median-of-three strategy can be used to improve performance.

## **Data Distribution Issues:**

The performance of Quick Sort depends on how the advertisement scores are distributed.

- Randomly distributed scores: Usually gives good performance.
- Already sorted data: Can cause poor performance with an unsuitable pivot.
- Many duplicate scores: Can create unnecessary partitions if not handled properly.
- Highly unbalanced data: Increases the number of recursive operations.

## **Performance Analysis:**

| Case         | Time Complexity |
|--------------|-----------------|
| Best Case    | $O(n \log n)$   |
| Average Case | $O(n \log n)$   |
| Worst Case   | $O(n^2)$        |

The average performance is efficient for ranking a large number of advertisements.

## **Optimizations:**

The following techniques can improve Quick Sort:

- Use random pivot selection to reduce the chance of worst-case behavior.

- Use median-of-three pivot selection for better partitioning.
- Handle duplicate scores efficiently.
- Use insertion sort for very small partitions.
- Avoid excessively deep recursion.

## **Conclusion:**

Quick Sort provides an efficient divide-and-conquer approach for online advertisement ranking. It can quickly arrange advertisements according to their ranking scores. Proper pivot selection and handling of duplicate or unevenly distributed data are important for maintaining good performance, especially when the number of advertisements is large.